SHIHAO ZHUO

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EDUCATION

University of Birmingham Birmingham, UK
MSc, Artificial Intelligence and Machine Learning 2026.09 - present

Ningbo University Ningbo, China
BEng, Computer Science and Technology 2018.09 - 2024.06

- The six-year span includes two years of compulsory military service (2019.09–2021.09).

EXPERIENCE

HKUST (Guangzhou), Research Assistant, Data-Centric AI 2026.03-2026.07

- Responsible for method design, experiments and paper writing on data curation; outputs are listed under Research Projects and Publications.

Shanghai Jiao Tong University, Research Assistant 2024.06-2024.12

- Independently built and maintained the research group website (`aipsybrain.com`); developed a Python GUI application for cognitive assessment tasks, supporting memory and cognition scales, deployed for hospital-side data collection.

Anlai Scientific Instruments (Ningbo) Co., Ltd., Intern Assistant 2024.07-2024.10

- Applied YOLOv8 to a mouse scratching-behaviour analysis project: detection data processing, annotation auditing and model tuning.

Ningbo Fanxing Consulting Co., Ltd., Development Engineer (Intern) 2023.04-2023.07

- Built a college-application recommendation system as a WeChat Mini Program, using WeChat Cloud's database to simplify storage.

RESEARCH PROJECTS

- **RMD^2: a low-cost curation operator for large data pipelines** (HKUST Guangzhou) 2026.03-2026.07
  Class-conditional statistics are fitted once on a small trusted reference set, after which scoring a candidate is one encoder pass and one quadratic form, with no task-proxy model trained anywhere in the loop. Label consistency is judged by an inter-class margin rather than an absolute distance, and the same fitted statistics serve both noisy-sample curation and generation guidance. Built the two-modality pipeline (DINOv2, BGE, ResNet-9, BERT-base, Qwen2.5-1.5B with LoRA/DPO) and benchmarked it on 4×RTX A5000 against both proxy-based data-attribution methods and proxy-free feature-space selectors.
- **Multi-Center MRI Image Classification Based on Deep Learning** (Ningbo University) 2024.01-2024.05
  To address the loss of generalisation caused by multi-center data heterogeneity, diffusion models and domain-transfer techniques align unseen-center data with known-center data, improving cross-center transferability of the classifier. The method is now a granted invention patent (first inventor).
- **Accurate Rib Fracture Diagnosis with Lightweight Deep Models** (Ningbo University) 2022.03-2023.03
  A lightweight network for chest CT that localises and classifies rib fracture regions and generates diagnostic reports, raising an alert when the radiologist's reading disagrees with the model prediction, so as to reduce missed and incorrect diagnoses.

PUBLICATIONS AND PATENTS

- **RMD^2: One Geometry, Two Roles in Data Curation and Generation Guidance** **In preparation, PVLDB**
  Co-first author.
- **Multi-Center MRI Image Classification Method**, invention patent 2024.08
  Granted, CN117173489B; **first inventor**.
- **A Sequential Multi-Classification Method for sMRI Images**, invention patent 2024.08
  Granted, CN116740463B; fifth inventor.

- **sMRI Classification with a High-Resolution Complementary-Attention UNet**, invention patent Granted, CN114332535B; fifth inventor. 2022.07
- **Fanxing Sanyi Information Consulting Mini Program**, software copyright 2023.05

Awards and Honours

- Outstanding Undergraduate Student, Ningbo University 2022–2023
- Third-Class University Scholarship, Ningbo University 2021–2022
- Outstanding Student Cadre, Ningbo University 2021–2022

Technical Skills

- **Programming**: Python (proficient), C/C++, SQL, JavaScript/HTML/CSS, LaTeX
- **Machine learning**: PyTorch, HuggingFace Transformers / PEFT (LoRA) / TRL (DPO), scikit-learn, NumPy/SciPy, YOLO, multi-GPU training
- **Tools and platforms**: Linux, macOS, Git, MongoDB, openGauss, WordPress
- **Languages**: Chinese (native), English (MSc taught in English; research written in English)